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ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of the Course: B. Tech (EE)

Semester: VII

Name of the Paper – Utilization of Electrical Energy

Paper Code: TEE 706

Time: 3Hours

Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	CO1
(a)	Discuss the Laws of illumination along with suitable diagrams.	
(b)	A school classroom, 7 m × 10 m × 4 m high is to be illuminated to 135 lm/m ² on the working plane. If the coefficient of utilization is 0.45 and the sources give 13 lumens per watt, work out the total wattage required, assuming a depreciation factor of 0.8. Sketch roughly the plan of the room, showing suitable positions for fittings, giving reasons for the positions chosen.	
(c)	A lamp of 100 candela is placed 1 m below a plane mirror which reflects 90% of light falling on it. The lamp is hung 4 m above ground. Find the illumination at a point on the ground 3 m away from the point vertically below the lamp.	
Q2	(20 marks)	CO2
(a)	Discuss the various modes of heat dissipation.	
(b)	A slab of insulating material 150 cm ² in area and 1 cm thick is to be heated by dielectric heating. The power required is 400 W at 30 MHz. Material has relative permittivity of 5 and p.f. of 0.06. Determine the necessary voltage. Absolute permittivity = 8.854×10^{-12} F/m. What are the applications of dielectric heating?	
(c)	A 30-kW, 3-φ, 400-V resistance oven is to employ nickel-chrome strip 0.254 mm thick for the three star-connected heating elements. If the wire temperature is to be 1,100°C and that of the charge to be 700°C, estimate a suitable width for the strip. Assume emissivity = 0.9 and radiating efficiency to be 0.5 and resistivity of the strip material is $101.6 \times 10^{-8} \Omega$ - m. What would be the temperature of the wire if the charge were cold?	

Q3	(20 marks)	CO3
(a)	Differentiate between fusion and non-fusion welding.	
(b)	Discuss the importance of Arc Welding Machines. What are the advantages of coated electrodes?	
(c)	Briefly explain the different methods of electric welding and state their relative merits.	
Q4	(20 marks)	CO4
(a)	How can the efficient utilization of electrical energy contribute to sustainability and reduce environmental impact in modern industries?	
(b)	What are the advantages of Electric Traction? Discuss the General Features of Traction Motor.	
(c)	A suburban train runs with an average speed of 36 km/h between two stations 2 km apart. Values of acceleration and retardation are 1.8 km/h/s and 3.6 km/h/s compute the maximum speed of the train assuming trapezoidal speed/time curve.	
Q5	(20 marks)	CO5
(a)	Discuss why a D.C. series motor is ideally suited for traction services.	
(b)	Discuss the Typical speed/time curve for electric trains.	
(c)	Explain the terms tractive effort, coefficient of adhesion, train resistance and specific energy consumption of train.	